

Daniel Traian Pele

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ABOUT ME

More recently, I study how large language models and time-series foundation models can be used to forecast risk, partly in joint work with my PhD students and the partners of the MSCA Doctoral Network on Digital Finance.

WORK EXPERIENCE

2022 – CURRENT Bucharest, Romania

Senior Researcher I (CS I)

Institute for Economic Forecasting (IPE), Romanian Academy

Head of the Department of Computer-Assisted Economic Research.

2020 – CURRENT Bucharest, Romania

Professor

Bucharest University of Economic Studies (ASE), Department of Statistics and Econometrics

Courses: Statistics, Econometrics, Time Series, Financial Markets Statistics, Advanced Time-Series Modelling, Financial Markets Modelling; supervision of bachelor and master theses.

2019 – CURRENT Bucharest, Romania

PhD supervisor

Doctoral School of Cybernetics and Economic Statistics, ASE

2018 – 2019 Berlin, Germany

Researcher, IRTG 1792 "High Dimensional Nonstationary Time Series"

Humboldt-Universität zu Berlin, with Prof. Wolfgang Karl Härdle

2016 – 2020

Statistical expert

European Investment Bank, European Commission, National School of Political and Administrative Studies

Sampling methodology for the ex-ante control of public procurement (EIB); minimum-wage setting mechanism for Romania (European Commission, INCSMPS); cost standards for local public services (SIPOCA 9).

2014 Berlin, Germany

Visiting researcher

Research Data Center, Humboldt-Universität zu Berlin, with Prof. Wolfgang Karl Härdle (Quantlet platform)

Development of the Quantlet platform in SAS for Statistics of Financial Markets.

2014 – 2024 Bucharest, Romania

Statistical expert

The World Bank: tax framework, public procurement, disaster-risk financing, governance scorecards

Tax framework under Romania's Recovery and Resilience Plan; public procurement strategy; disaster-risk financing and the Bucharest seismic-risk tool; Country Climate and Development Report for Azerbaijan; Moldova Governance Scorecard; Romania Strategy Unit; sampling methodology for the Human Capital Operational Programme.

2014 – 2015

Statistical expert, Delivery Unit

Prime Minister's Chancellery, Romania (World Bank project)

Statistical analysis, monitoring and evaluation, government dashboard, risk model for VAT refunds; analyses of public procurement, youth employment, tax administration and energy.

2007 – 2021 Bucharest, Romania

Statistician

Ipsos Interactive Services, Bucharest: statistical modelling

2007 Bucharest, Romania

Senior adviser

National Institute of Statistics, Bucharest: statistical analysis and sampling

2005 – 2007 Bucharest, Romania

Statistician

MEMRB RTS, Bucharest: retail-audit analysis of the FMCG market and marketing-mix models

2004 – 2005 Bucharest, Romania

Risk analyst

Credisson International, Bucharest: scoring models

2001 – 2020 Bucharest, Romania

Teaching assistant (2001), assistant (2003), lecturer (2007), associate professor (2013)

ASE, Department of Statistics and Econometrics

EDUCATION AND TRAINING

2019 Bucharest, Romania

Habilitation, Cybernetics and Statistics

ASE. Thesis: Statistical methods for financial markets

2011 Reading, United Kingdom

Postdoctoral fellowship

ICMA Centre, University of Reading, with Emese Lazar: uncertainty in capital markets via information entropy

2010 – 2013 Bucharest, Romania

Postdoctoral researcher

ASE: predictability of financial crises

Topic: predictability of financial crises, with applications to the stock market. Advisers: Nicolae Dardac, Gabriela Anghelache.

2007 Bucharest, Romania

PhD in Economics (Cybernetics and Statistics)

ASE. Advisor: Vergil Voineagu

Thesis: Statistical methods for the analysis of the stock market.

2002 Bucharest, Romania

MSc in Statistics (Stochastic Processes and Theoretical Statistics)

University of Bucharest, Faculty of Mathematics. Advisor: Monica Dumitrescu

Dissertation: Repeated surveys through linear time-series models.

2000 Bucharest, Romania

BSc in Mathematics (Advanced Studies)

University of Bucharest, Faculty of Mathematics. Advisor: Anton Ștefănescu

Thesis: Competitive equilibrium in the Arrow–Debreu model.

1992 – 1996 Corabia, Romania

High school

Alexandru Ioan Cuza National College (then Theoretical High School), Corabia, Olt County

LANGUAGE SKILLS

Mother tongue: Romanian

Other languages:

	Listening	Reading	Spoken interaction	Spoken production	Writing
English	B2–C1: certified overall level (PROSPER-ASE certificate no. 8472, 30.06.2022)				
French	B2	B2	B2	B2	B2

Levels: A1 and A2 basic user; B1 and B2 independent user; C1 and C2 proficient user (Common European Framework of Reference for Languages).

DIGITAL SKILLS

Statistical and econometric software: SAS, R, MATLAB, SPSS, EViews, Stata. Programming and data science: Python (pandas, statsmodels, scikit-learn, PyTorch), Jupyter and Google Colab. Scientific writing and reproducible research: LaTeX and Beamer, Git and GitHub, Quantlet. Microsoft Office.

Self-assessment (Europass digital competence grid): proficient user in information processing, communication, content creation, safety and problem solving.

ORGANISATIONAL SKILLS

- Founded the SAS Centre of Excellence at ASE Bucharest.
- Organised the AIDA Conference 2026 (AI and Digital Assets in Energy and Finance, Bucharest) and the workshop “AI, Digital Assets and the Future of Energy Finance” (2024); minitrack chair at ICBE.
- Project director for ASE in the MSCA Doctoral Network DIGITAL and co-leader of Work Package 4.

PROJECTS

2024 – CURRENT Bucharest, Romania

DIGITAL: MSCA Industrial Doctoral Network on Digital Finance

Horizon Europe, No. 101119635

Project director for ASE and co-leader of Work Package 4, "Driving digital innovation with Blockchain applications"; industrial doctoral training in AI for finance.

2023 – 2026

IDA: Institute for Digital Assets

PNRR, contract CN760046/23.05.2023

Deputy Director. Research: foundation models and LLMs for risk forecasting (LLM-VaR, LLM-ES), machine learning for digital assets (early warning for “zombie” assets, liquidity forecasting).

2023 – 2026

AI4EFin: Artificial Intelligence for Energy Finance

PNRR, contract CN760048/23.05.2023

Researcher; led the research line on forecasting, explainable AI and statistical calibration and coordinated the PhD students in the team: energy-price forecasting, early warning for price spikes, conformal calibration.

2020 – 2024

COST Action CA19130: Fintech and AI in Finance

Management Committee member for Romania; trainer in the Action’s summer schools.

2019 – 2021

FIN-TECH: Horizon 2020

Horizon 2020, No. 825215

Financial supervision and technology compliance training programme: an EU-wide risk-management platform.

2018 – 2019 Berlin, Germany

IRTG 1792: High Dimensional Nonstationary Time Series

Humboldt-Universität zu Berlin

Researcher on the genus–differentia classification of cryptocurrencies, with Prof. Wolfgang Karl Härdle.

2021

Unified methodology for damage assessment in emergencies and disasters

World Bank and the General Inspectorate for Emergency Situations, contract no. 111823/20.08.2021

2012 – 2013

Pre-financing rate of the EU structural instruments

ACIS, contract no. 3/23/57/26.09.2012: econometric model of the impact of pre-financing on the absorption rate

2011 – 2012

Institutional capacity of the National Forecasting Commission (SMIS 27153)

Studies on the contribution of SMEs to economic growth and on the private sector in research and innovation: model of the impact of R&D spending on growth

2011 – 2012 Bucharest, Romania

Professionalism in the county statistics offices (SMIS 15966)

ASE with the National Institute of Statistics

2009 – 2011

CNCSIS IDEI research grants ID_1820 and ID_183

Team member

2008 – 2011

Quality in education (CALE), POSDRU

Econometric model forecasting labour demand in construction under the macroeconomic and crisis context

2006

CEEX research project

Team member

2002 – 2006

CERES research projects

Team member

NETWORKS AND MEMBERSHIPS

- Associate Editor, *Digital Finance* (Springer)
- Member, Romanian National Statistical Council (2023–)
- Board member for Romania, European Courses in Advanced Statistics (ECAS, 2020–)
- Management Committee member, COST Action CA19130 (2020–2024)
- Minitrack chair, International Conference on Business Excellence (ICBE 2023, “Applying ML and AI to Finance, Business and Society”; ICBE 2025, IDA minitrack)

HONOURS AND AWARDS

- Editor of Distinction Award (Author Service Award 2026), Springer Nature, for editorial work at Digital Finance
- Best Paper Award, International Conference on Business Excellence (ICBE) 2026, for “A Multimodal Vision-Language Framework for Financial Anomaly Detection”, with S.-L. Jheng, R. Tak and Ș. Găman

- Honorary Citizen of Gura Padinii commune, Olt County (2024)<img src="assets/a_gura_1.jpg" alt="" width="480" height="640" loading="lazy" style="aspect-ratio:3/4"><img src="assets/a_gura_2.jpg" alt="" width="480" height="640" loading="lazy" style="aspect-ratio:3/4"><img src="assets/a_gura_3.jpg" alt="" width="480" height="640" loading="lazy" style="aspect-ratio:3/4">
- Best Paper Award, International Conference on Business Excellence (ICBE) 2019 (Journal of Economic Forecasting Award), with W. K. Härdle and N. Wesselhöfft
- "Bologna Professor" diploma (2010)
- Diplomas of excellence, Romanian Statistical Society (2009, 2010, 2012)

PUBLICATIONS

99 publications, of which 45 journal articles; full list with links and BibTeX: https://danpele.github.io/publications.html